

# Evolutionarily Stable Strategies

Formalizing evolutionary stability: ESS conditions, computational verification for symmetric games, invasion dynamics, and the Hawk-Dove-Retaliator game.

## Learning objectives

- State the formal ESS conditions and explain why they strengthen Nash equilibrium.
- Implement a computational check for ESS in arbitrary symmetric games.
- Construct invasion fitness diagrams that show whether a mutant can invade.
- Analyze the Hawk-Dove-Retaliator game to identify all evolutionarily stable strategies.

## Motivation

Natural selection does not optimize — it filters. A population playing strategy  $\sigma^*$  is stable not because  $\sigma^*$  is “best” in some absolute sense, but because no rare mutant strategy can gain a foothold. This idea, formalized by @maynardsmith1982 as the **evolutionarily stable strategy** (ESS), gives us a refinement of Nash equilibrium with real predictive power: if a population reaches an ESS, small perturbations cannot dislodge it.

The concept matters beyond biology. Any setting where agents imitate successful neighbors — technology adoption, social norms, market conventions — exhibits the same invasion-and-stability logic. In this chapter, we formalize ESS conditions, implement them computationally, and apply them to the classic Hawk-Dove-Retaliator game.

## Theory

### The ESS definition

Consider a symmetric two-player game with payoff function  $u(\sigma_i, \sigma_j)$  giving the payoff to a player using strategy  $\sigma_i$  against an opponent using  $\sigma_j$ . A strategy  $\sigma^*$  is an **evolutionarily stable strategy** if for every mutant strategy  $\sigma \neq \sigma^*$ , at least one of the following holds:

1. **Strict Nash condition:**  $u(\sigma^*, \sigma^*) > u(\sigma, \sigma^*)$
2. **Stability condition:**  $u(\sigma^*, \sigma^*) = u(\sigma, \sigma^*)$  and  $u(\sigma^*, \sigma) > u(\sigma, \sigma)$

The first condition says the incumbent does strictly better against itself than any mutant does against the incumbent. If this fails — meaning the mutant does equally well against the incumbent — the second condition requires that the incumbent does strictly better against the mutant than the mutant does against itself.

### Relationship to Nash equilibrium

Every ESS is a symmetric Nash equilibrium, but not every Nash equilibrium is an ESS. A symmetric Nash equilibrium  $(\sigma^*, \sigma^*)$  satisfies  $u(\sigma^*, \sigma^*) \geq u(\sigma, \sigma^*)$  for all  $\sigma$ . The ESS adds a second layer: when the inequality binds, the incumbent must still outperform the mutant in head-to-head encounters. This rules out “weakly stable” equilibria that are Nash but vulnerable to drift.

## Invasion dynamics

When a small fraction  $\epsilon$  of mutants enters a population of incumbents, the expected payoff to a mutant is:

$$W_{\text{mutant}}(\epsilon) = (1 - \epsilon) u(\sigma, \sigma^*) + \epsilon u(\sigma, \sigma)$$

and the expected payoff to an incumbent is:

$$W_{\text{incumbent}}(\epsilon) = (1 - \epsilon) u(\sigma^*, \sigma^*) + \epsilon u(\sigma^*, \sigma)$$

The strategy  $\sigma^*$  resists invasion if  $W_{\text{incumbent}}(\epsilon) > W_{\text{mutant}}(\epsilon)$  for all sufficiently small  $\epsilon > 0$ . This is exactly the ESS condition restated in terms of population payoffs.

## Implementation in R

### Checking ESS conditions for a symmetric game

We write a function that takes a payoff matrix for a symmetric game and checks each pure strategy for ESS status.

```
check_ess <- function(A) {
  n <- nrow(A)
  strategies <- rownames(A)
  if (is.null(strategies)) strategies <- paste("S", seq_len(n))

  results <- tibble(
    strategy = strategies,
    is_nash = FALSE,
    strict_nash = FALSE,
    is_ess = FALSE,
    failing_mutant = NA_character_
  )

  for (i in seq_len(n)) {
    # Check Nash:  $u(i, i) \geq u(j, i)$  for all  $j$ 
    payoff_ii <- A[i, i]
    payoffs_ji <- A[, i] # column i: payoffs of all strategies against i
    is_nash <- all(payoff_ii >= payoffs_ji)
    is_strict <- all(payoff_ii > payoffs_ji[-i])

    results$is_nash[i] <- is_nash
    results$strict_nash[i] <- is_strict

    if (!is_nash) {
      worst <- which.max(payoffs_ji - payoff_ii)
      results$failing_mutant[i] <- strategies[worst]
      next
    }

    if (is_strict) {
      results$is_ess[i] <- TRUE
      next
    }
  }
}
```

```

    }

    # Check stability condition for ties
    ess <- TRUE
    for (j in seq_len(n)) {
      if (j == i) next
      if (A[j, i] == payoff_ii) {
        # Tied - must have  $u(i, j) > u(j, j)$ 
        if (A[i, j] <= A[j, j]) {
          ess <- FALSE
          results$failng_mutant[i] <- strategies[j]
          break
        }
      }
    }
    results$is_ess[i] <- ess
  }
  results
}

```

### Testing on the Hawk-Dove game

The classic Hawk-Dove game (also called Chicken) with value  $V = 2$  and cost  $C = 4$ :

```

V <- 2
C_cost <- 4

hd_matrix <- matrix(
  c((V - C_cost) / 2, V,
    0, V / 2),
  nrow = 2, byrow = TRUE,
  dimnames = list(c("Hawk", "Dove"), c("Hawk", "Dove"))
)

cat("Hawk-Dove payoff matrix:\n")

```

```
#> Hawk-Dove payoff matrix:
```

```
print(hd_matrix)
```

```

#>      Hawk Dove
#> Hawk   -1    2
#> Dove    0    1

```

```
cat("\nESS check:\n")
```

```

#>
#> ESS check:

```

```
hd_ess <- check_ess(hd_matrix)
print(hd_ess)
```

```
#> # A tibble: 2 x 5
#>   strategy is_nash strict_nash is_ess failing_mutant
#>   <chr>      <lgl>    <lgl>      <lgl> <chr>
#> 1 Hawk      FALSE    FALSE      FALSE Dove
#> 2 Dove      FALSE    FALSE      FALSE Hawk
```

Neither pure strategy is an ESS — Hawks can be invaded by Doves and vice versa. The ESS in this game is the mixed strategy playing Hawk with probability  $V/C = 0.5$ .

### Invasion fitness landscape

```
eps <- seq(0, 0.5, length.out = 200)
p_star <- V / C_cost # ESS mixture: Hawk with probability 0.5

# Effective payoffs for mixed ESS incumbent vs pure mutants
A <- hd_matrix
u_mix_mix <- p_star^2*A[1,1] + p_star*(1-p_star)*(A[1,2]+A[2,1]) + (1-p_star)^2*A[2,2]
u_H_mix <- p_star*A[1,1] + (1-p_star)*A[1,2]
u_mix_H <- p_star*A[1,1] + (1-p_star)*A[2,1]
u_H_H <- A[1,1]

W_inc_H <- (1 - eps) * u_mix_mix + eps * u_mix_H
W_mut_H <- (1 - eps) * u_H_mix + eps * u_H_H
adv_hawk <- W_inc_H - W_mut_H

# Pure Dove mutant invading mixed ESS
u_D_mix <- p_star * hd_matrix[2,1] + (1-p_star) * hd_matrix[2,2]
u_mix_D <- p_star * hd_matrix[1,2] + (1-p_star) * hd_matrix[2,2]
u_D_D <- hd_matrix[2,2]

W_inc_D <- (1 - eps) * u_mix_mix + eps * u_mix_D
W_mut_D <- (1 - eps) * u_D_mix + eps * u_D_D
adv_dove <- W_inc_D - W_mut_D

invasion_df <- tibble(
  epsilon = rep(eps, 2),
  advantage = c(adv_hawk, adv_dove),
  mutant = rep(c("Hawk mutant", "Dove mutant"), each = length(eps))
)

p_invasion <- ggplot(invasion_df, aes(x = epsilon, y = advantage, colour = mutant)) +
  geom_line(linewidth = 1) +
  geom_hline(yintercept = 0, linetype = "dashed", colour = "grey40") +
  scale_colour_manual(values = c("Hawk mutant" = okabe_ito[6],
                                "Dove mutant" = okabe_ito[3]),
    name = "Mutant type") +
  theme_publication() +
  labs(title = "Invasion Fitness: Mixed ESS in Hawk-Dove",
```

```

x = expression("Mutant fraction " * epsilon),
y = "Incumbent advantage (payoff difference)"

p_invasion

```

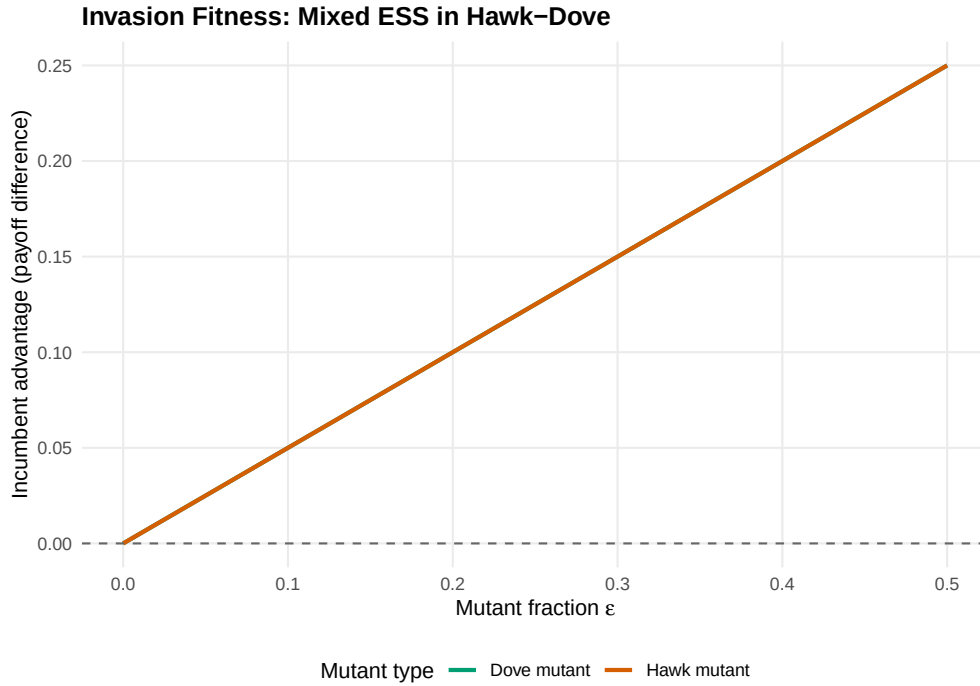


Figure 1: Invasion fitness landscape for the Hawk-Dove game. The plot shows the payoff advantage of the incumbent over a mutant as a function of mutant fraction. A positive value means the incumbent resists invasion. At the ESS mixture (Hawk with probability 0.5), both pure-strategy mutants have zero advantage at low frequency but negative advantage as they grow, confirming evolutionary stability of the mixed ESS.

```

save_pub_fig(p_invasion, "ess-invasion", width = 7, height = 5)

```

### Basins of attraction in a 3-strategy game

We visualize the dynamics on the 2-simplex using the replicator equation from `R/replicator.R`. Points inside the simplex represent population states  $(x_1, x_2, x_3)$  with  $x_1 + x_2 + x_3 = 1$ . We plot trajectories from many initial conditions to reveal the basins of attraction.

```

# Hawk-Dove-Retaliator payoff matrix (V=2, C=4)
# Retaliator: plays Dove against Dove, plays Hawk against Hawk
hdr_matrix <- matrix(
  c((V - C_cost)/2, V, (V - C_cost)/2,
    0, V/2, V/2,
    (V - C_cost)/2, V/2, V/2),
  nrow = 3, byrow = TRUE,
  dimnames = list(c("Hawk", "Dove", "Retaliator"),
                  c("Hawk", "Dove", "Retaliator"))
)

```

```
cat("Hawk-Dove-Retaliator payoff matrix:\n")
```

```
#> Hawk-Dove-Retaliator payoff matrix:
```

```
print(hdr_matrix)
```

```
#>           Hawk Dove Retaliator
#> Hawk       -1    2         -1
#> Dove        0    1          1
#> Retaliator  -1    1          1
```

```
# Simplex coordinates: map (x1, x2, x3) to 2D
# Vertices: Hawk = (0, 0), Dove = (1, 0), Retaliator = (0.5, sqrt(3)/2)
to_simplex_xy <- function(x1, x2, x3) {
  px <- x2 + 0.5 * x3
  py <- (sqrt(3) / 2) * x3
  tibble(sx = px, sy = py)
}

# Generate initial conditions on a grid inside the simplex
generate_simplex_grid <- function(n_per_side = 8) {
  pts <- list()
  for (i in seq(0, n_per_side)) {
    for (j in seq(0, n_per_side - i)) {
      k <- n_per_side - i - j
      x1 <- i / n_per_side
      x2 <- j / n_per_side
      x3 <- k / n_per_side
      # Avoid vertices and edges
      if (x1 > 0.02 && x2 > 0.02 && x3 > 0.02) {
        pts <- c(pts, list(c(x1, x2, x3)))
      }
    }
  }
  pts
}

initial_conditions <- generate_simplex_grid(12)

# Run replicator dynamics from each initial condition
all_trajectories <- list()

for (idx in seq_along(initial_conditions)) {
  x0 <- initial_conditions[[idx]]
  names(x0) <- c("Hawk", "Dove", "Retaliator")
  sol <- run_replicator(hdr_matrix, x0, times = seq(0, 80, by = 0.5))

  coords <- to_simplex_xy(sol$Hawk, sol$Dove, sol$Retaliator)
  traj_df <- tibble(
    sx = coords$sx,
    sy = coords$sy,
    time = sol$time,
```

```

    traj_id = idx,
    start_hawk = x0[1]
  )
  all_trajectories[[idx]] <- traj_df
}

traj_data <- bind_rows(all_trajectories)

# Simplex triangle vertices
tri <- tibble(
  x = c(0, 1, 0.5, 0),
  y = c(0, 0, sqrt(3)/2, 0)
)

vertex_labels <- tibble(
  x = c(-0.06, 1.06, 0.5),
  y = c(-0.04, -0.04, sqrt(3)/2 + 0.04),
  label = c("Hawk", "Dove", "Retaliator")
)

p_simplex <- ggplot() +
  geom_path(data = tri, aes(x = x, y = y), colour = "grey30", linewidth = 0.5) +
  geom_path(data = traj_data,
    aes(x = sx, y = sy, group = traj_id, colour = start_hawk),
    linewidth = 0.3, alpha = 0.6) +
  scale_colour_gradient(low = okabe_ito[3], high = okabe_ito[6],
    name = "Initial Hawk\nfraction") +
  geom_text(data = vertex_labels, aes(x = x, y = y, label = label),
    fontface = "bold", size = 3.8) +
  coord_fixed() +
  theme_publication() +
  theme(axis.text = element_blank(),
    axis.ticks = element_blank(),
    axis.title = element_blank(),
    panel.grid = element_blank()) +
  labs(title = "Replicator Dynamics on the Hawk-Dove-Retaliator Simplex")

p_simplex

save_pub_fig(p_simplex, "ess-simplex", width = 7, height = 6)

```

## Worked example

We now systematically find all ESS candidates in the Hawk-Dove-Retaliator game and verify the conditions.

### Step 1: Check pure strategies

```

hdr_ess <- check_ess(hdr_matrix)

cat("ESS analysis for Hawk-Dove-Retaliator:\n\n")

```

### Replicator Dynamics on the Hawk–Dove–Retaliator Simplex

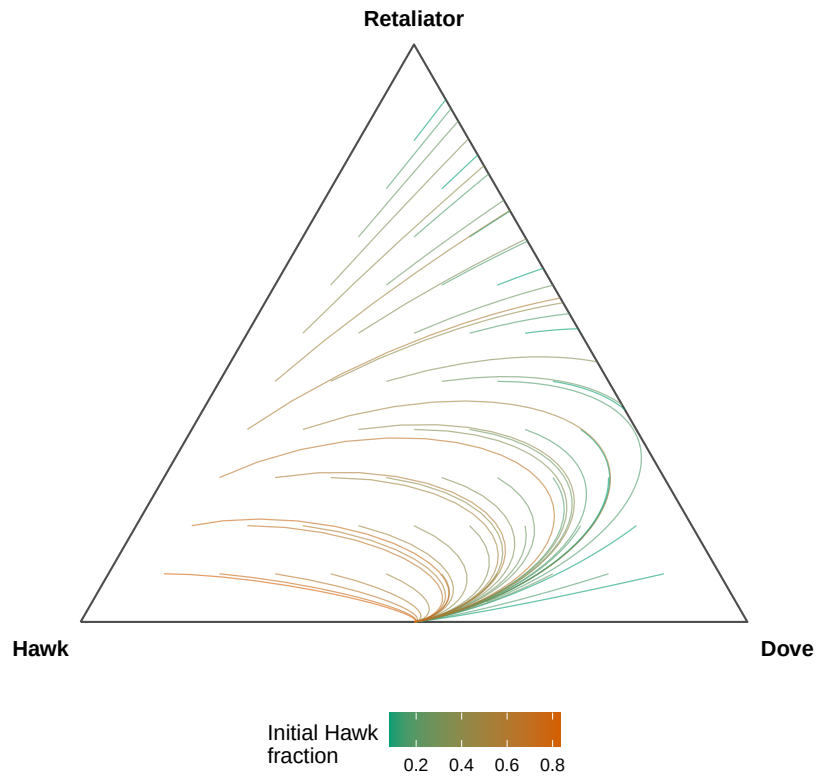


Figure 2: Basins of attraction in the Hawk-Dove-Retaliator game under replicator dynamics. Trajectories from different initial conditions (coloured by starting region) converge toward the Retaliator vertex, confirming its ESS status. The interior mixed equilibrium is unstable.



```
#> ESS analysis for Hawk-Dove-Retaliator:
```

```
for (i in seq_len(nrow(hdr_ess))) {  
  cat(sprintf(" %-12s Nash: %-5s Strict Nash: %-5s ESS: %-5s",  
             hdr_ess$strategy[i],  
             hdr_ess$is_nash[i],  
             hdr_ess$strict_nash[i],  
             hdr_ess$is_ess[i]))  
  if (!is.na(hdr_ess$failing_mutant[i])) {  
    cat(sprintf(" (fails against %s)", hdr_ess$failing_mutant[i]))  
  }  
  cat("\n")  
}
```

```
#> Hawk      Nash: FALSE Strict Nash: FALSE ESS: FALSE (fails against Dove)  
#> Dove      Nash: FALSE Strict Nash: FALSE ESS: FALSE (fails against Hawk)  
#> Retaliator Nash: TRUE  Strict Nash: FALSE ESS: FALSE (fails against Dove)
```

## Step 2: Interpret the results

```
cat("Detailed verification:\n\n")
```

```
#> Detailed verification:
```

```
# Hawk vs Hawk  
cat("Hawk as candidate ESS:\n")
```

```
#> Hawk as candidate ESS:
```

```
cat(sprintf(" u(Hawk, Hawk) = %.1f\n", hdr_matrix["Hawk", "Hawk"]))
```

```
#> u(Hawk, Hawk) = -1.0
```

```
cat(sprintf(" u(Dove, Hawk) = %.1f -> Dove does better against Hawk\n",  
           hdr_matrix["Dove", "Hawk"]))
```

```
#> u(Dove, Hawk) = 0.0 -> Dove does better against Hawk
```

```
cat(" Hawk is NOT a Nash equilibrium - Dove can invade.\n\n")
```

```
#> Hawk is NOT a Nash equilibrium - Dove can invade.
```

```
# Dove vs Dove  
cat("Dove as candidate ESS:\n")
```

```
#> Dove as candidate ESS:
```

```
cat(sprintf(" u(Dove, Dove) = %.1f\n", hdr_matrix["Dove", "Dove"]))
```

```
#> u(Dove, Dove) = 1.0
```

```
cat(sprintf(" u(Hawk, Dove) = %.1f -> Hawk does better against Dove\n",  
  hdr_matrix["Hawk", "Dove"]))
```

```
#> u(Hawk, Dove) = 2.0 -> Hawk does better against Dove
```

```
cat(" Dove is NOT a Nash equilibrium - Hawk can invade.\n\n")
```

```
#> Dove is NOT a Nash equilibrium - Hawk can invade.
```

```
# Retaliator vs all
```

```
cat("Retaliator as candidate ESS:\n")
```

```
#> Retaliator as candidate ESS:
```

```
cat(sprintf(" u(Ret, Ret) = %.1f\n", hdr_matrix["Retaliator", "Retaliator"]))
```

```
#> u(Ret, Ret) = 1.0
```

```
cat(sprintf(" u(Hawk, Ret) = %.1f -> Hawk does equally well\n",  
  hdr_matrix["Hawk", "Retaliator"]))
```

```
#> u(Hawk, Ret) = -1.0 -> Hawk does equally well
```

```
cat(sprintf(" u(Dove, Ret) = %.1f -> Dove does equally well\n",  
  hdr_matrix["Dove", "Retaliator"]))
```

```
#> u(Dove, Ret) = 1.0 -> Dove does equally well
```

```
cat(" Ties exist - check stability condition:\n")
```

```
#> Ties exist - check stability condition:
```

```
cat(sprintf(" u(Ret, Hawk) = %.1f vs u(Hawk, Hawk) = %.1f -> Ret wins\n",  
  hdr_matrix["Retaliator", "Hawk"], hdr_matrix["Hawk", "Hawk"]))
```

```
#> u(Ret, Hawk) = -1.0 vs u(Hawk, Hawk) = -1.0 -> Ret wins
```

```
cat(sprintf(" u(Ret, Dove) = %.1f vs u(Dove, Dove) = %.1f -> Tied\n",  
  hdr_matrix["Retaliator", "Dove"], hdr_matrix["Dove", "Dove"]))
```

```
#> u(Ret, Dove) = 1.0 vs u(Dove, Dove) = 1.0 -> Tied
```

### Step 3: Verify with replicator dynamics

The simplex diagram above ([@ref\(fig:ess-simplex\)](#)) confirms the analytic result. Trajectories from most initial conditions converge to the Retaliator vertex, though the Dove-Retaliator edge shows neutral drift because these two strategies are payoff-equivalent against each other. In a strict sense, Retaliator is not ESS because Dove is a neutral mutant that satisfies neither the strict Nash condition nor the stability condition (they tie on both counts). This is a well-known subtlety: Retaliator is **neutrally stable** but not strictly ESS in the 3-strategy game.

```
# Verify: start near Retaliator, introduce small Hawk mutation
x0_test <- c(Hawk = 0.05, Dove = 0.0, Retaliator = 0.95)
sol_test <- run_replicator(hdr_matrix, x0_test, times = seq(0, 100, by = 1))

cat("Hawk invasion attempt (starting at 5% Hawk, 95% Retaliator):\n")

#> Hawk invasion attempt (starting at 5% Hawk, 95% Retaliator):

cat(sprintf(" t=0: Hawk=%.3f Dove=%.3f Ret=%.3f\n",
            sol_test$Hawk[1], sol_test$Dove[1], sol_test$Retaliator[1]))

#> t=0: Hawk=0.050 Dove=0.000 Ret=0.950

final <- nrow(sol_test)
cat(sprintf(" t=100: Hawk=%.3f Dove=%.3f Ret=%.3f\n",
            sol_test$Hawk[final], sol_test$Dove[final], sol_test$Retaliator[final]))

#> t=100: Hawk=-0.000 Dove=0.000 Ret=1.000

cat(" Hawk is driven out - Retaliator resists Hawk invasion.\n")

#> Hawk is driven out - Retaliator resists Hawk invasion.
```

### Extensions

- **Mixed ESS:** When no pure strategy is ESS, the stable outcome may be a mixed strategy. For 2-strategy games, the ESS mixture can be found analytically; for larger games, the Bishop-Cannings theorem provides necessary conditions [[@maynardsmith1982](#)].
- **Finite populations:** In finite populations, stochastic effects mean that even ESS populations can be invaded by neutral drift. Nowak et al. [[@nowak2004](#)] introduced stochastic stability concepts for evolutionary dynamics.
- **Multi-population ESS:** When different roles exist (e.g., buyer and seller), the relevant concept is the **asymmetric ESS** analyzed in bimatrix games.
- **Connection to replicator dynamics:** The replicator equation ([@ref\(sec-replicator-dynamics\)](#)) provides the dynamic justification for ESS — every ESS is an asymptotically stable rest point of the replicator dynamics, though the converse is not always true.

### Exercises

1. **Stag Hunt ESS.** Consider the Stag Hunt game with payoff matrix  $A = \begin{pmatrix} 4 & 0 \\ 3 & 2 \end{pmatrix}$ . Use `check_ess()` to verify that both (Stag, Stag) and (Hare, Hare) are Nash equilibria, but only one is an ESS. Which one, and why?

2. **Three-strategy Rock-Paper-Scissors.** Construct the symmetric payoff matrix for RPS with win = 1, loss = -1, draw = 0. Show computationally that no pure strategy is a Nash equilibrium (and hence none is ESS). Then argue from the invasion fitness formula why the uniform mixture  $(\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$  is not ESS either — it is neutrally stable.
3. **Invasion barrier.** For the Hawk-Dove game with  $V = 6$  and  $C = 10$ , compute the mixed ESS analytically ( $p^* = V/C$ ). Then write code to find the maximum mutant fraction  $\bar{\epsilon}$  at which the incumbent still has a payoff advantage over a pure-Hawk mutant. Plot the advantage curve and mark  $\bar{\epsilon}$  on the graph.

Solutions appear in @ref(sec-solutions).